BAO CAO THUC NGHIEM LAB01

Run ID: 20261105_013545_b7cbe4
Trang thai: SUCCESS
Thoi gian bat dau: 2026-11-05T01:35:45+07:00
Thoi gian ket thuc: 2026-11-05T01:36:33+07:00

1. TIEN TRINH THUC HIEN

STT	Buoc thuc hien	Trang thai
1	EDA	SUCCESS
2	PREPROCESSING	SUCCESS
3	TRAINING - Logistic Regression	SUCCESS
4	TRAINING - Naive Bayes	SUCCESS
5	TRAINING - SVM	SUCCESS
6	TRAINING	SUCCESS
7	EVALUATION - Logistic Regression - Train	SUCCESS
8	EVALUATION - Logistic Regression - Test	SUCCESS
9	EVALUATION - Naive Bayes - Train	SUCCESS
10	EVALUATION - Naive Bayes - Test	SUCCESS
11	EVALUATION - SVM - Train	SUCCESS
12	EVALUATION - SVM - Test	SUCCESS
13	EVALUATION	SUCCESS
14	PIPELINE	SUCCESS

2. KET QUA DANH GIA MO HINH

Run_ID	Mô hình	T■p d■ li■u	Accuracy	Precision	Recall	F1	TPR	FPR	AUC	TN	FP	FN	TP
20261105_013545_b7cbe4	Logistic Regression	Train	0.95354729729729 71	0.900783289817 2324	0.953038674033 1492	0.926174496644 2952	0.953038674033 1492	0.046228710462 2871	0.971636143821 1611	3136	152	68	1380
20261105_013545_b7cbe4	Logistic Regression	Test	0.931587837837 8378	0.846913580246 9135	0.947513812154 696	0.894393741851 3689	0.947513812154 696	0.075425790754 2579	0.959679262276 3506	760	62	19	343
20261105_013545_b7cbe4	Naive Bayes	Train	0.996199324324 3245	0.989726027397 2602	0.997928176795 58	0.993810178817 0564	0.997928176795 58	0.004562043795 6204	0.999959042424 4869	3273	15	3	1445
20261105_013545_b7cbe4	Naive Bayes	Test	0.972972972972 973	0.963483146067 4156	0.947513812154 696	0.955431754874 6516	0.947513812154 696	0.015815085158 1508	0.993547606565 3104	809	13	19	343

Run_ID	Mô hình	T■p d■ li■u	Accuracy	Precision	Recall	F1	TPR	FPR	AUC	TN	FP	FN	TP
20261105_013545_b7cbe4	SVM	Train	0.71938344594594459	0.88888888888888888	0.09392265193370166	0.16989381636472	0.09392265193370166	0.00517031630170031	0.9747585813472 06	3271	17	1312	136
20261105_013545_b7cbe4	SVM	Test	0.71875	0.89189189189189 19	0.09116022099447 51	0.16541353383458 64	0.09116022099447 51	0.00486618004866 18	0.9647269158903 632	818	4	329	33

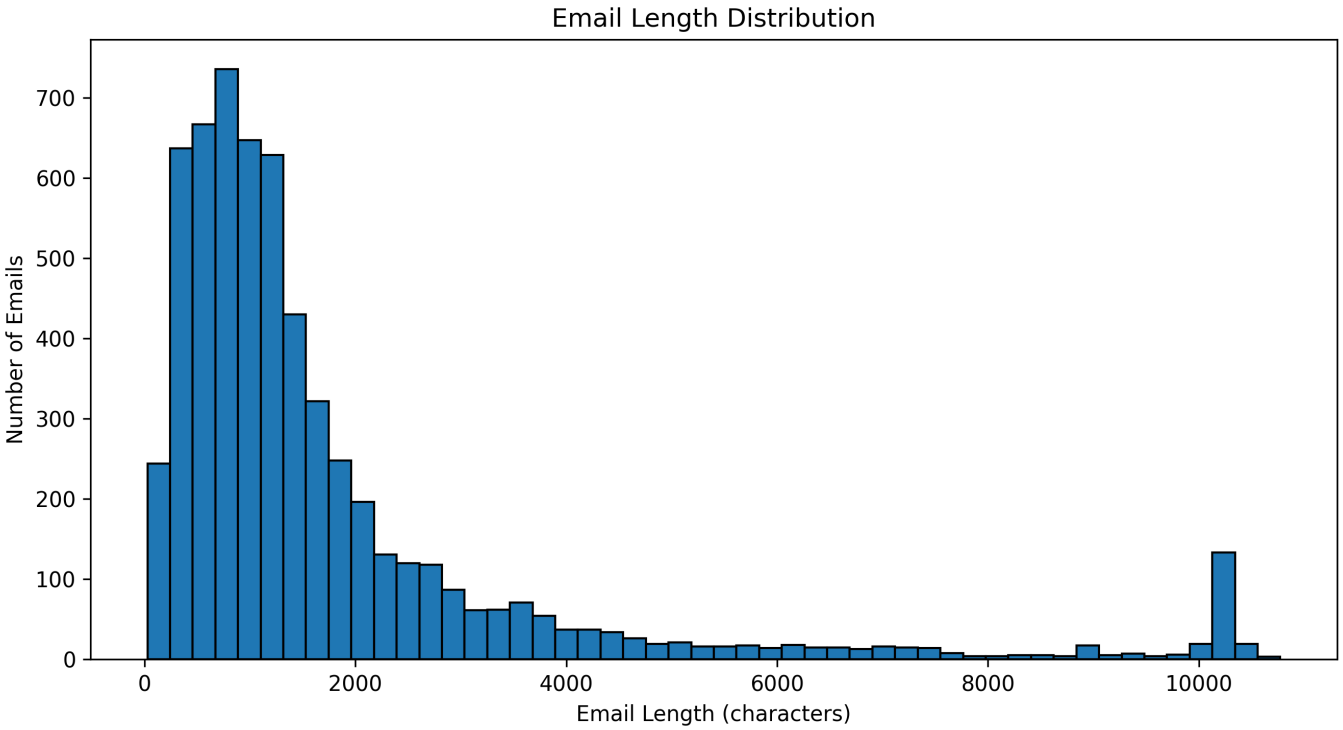
3. MO HINH TOT NHAT

Mo hinh: Naive Bayes
F1-score: 0.9554317548746516

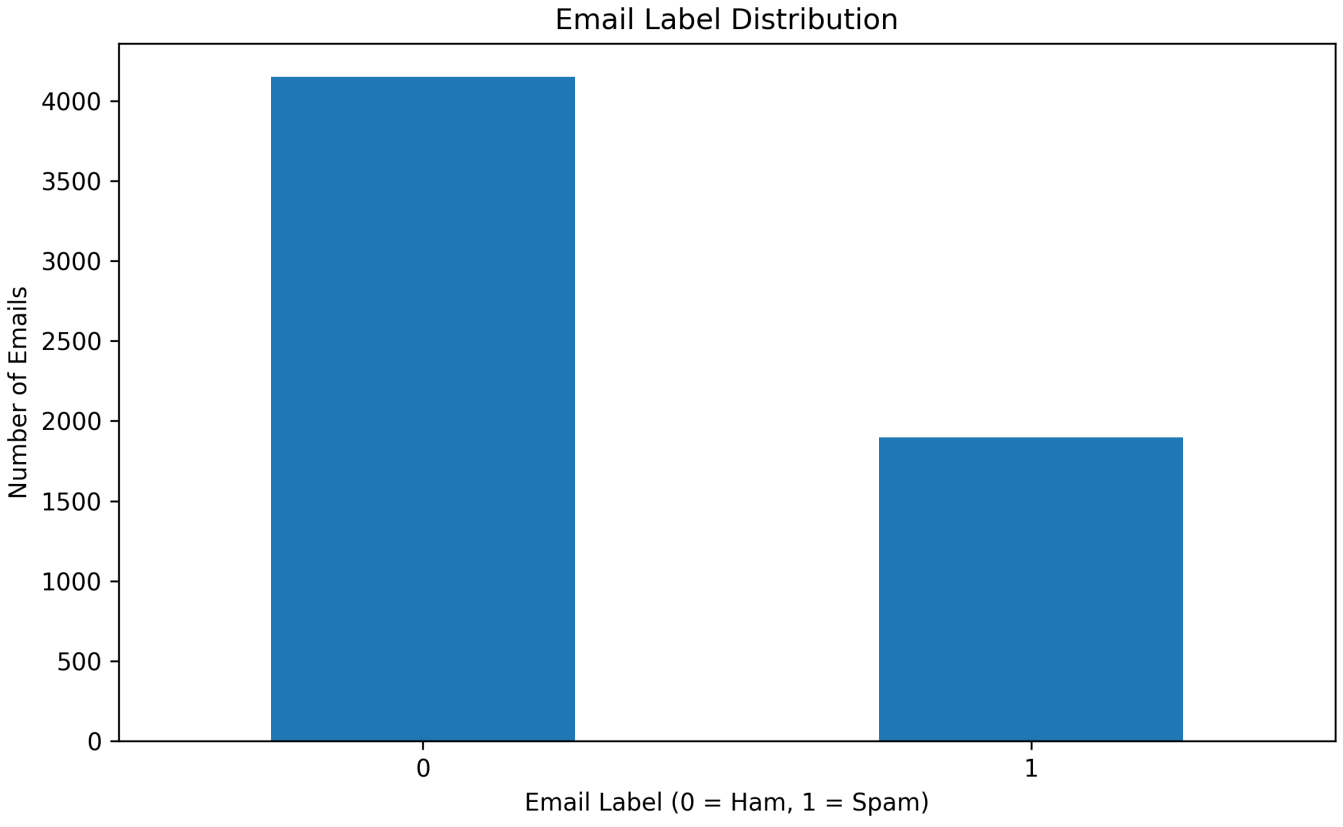
4. BIEU DO KET QUA

EDA

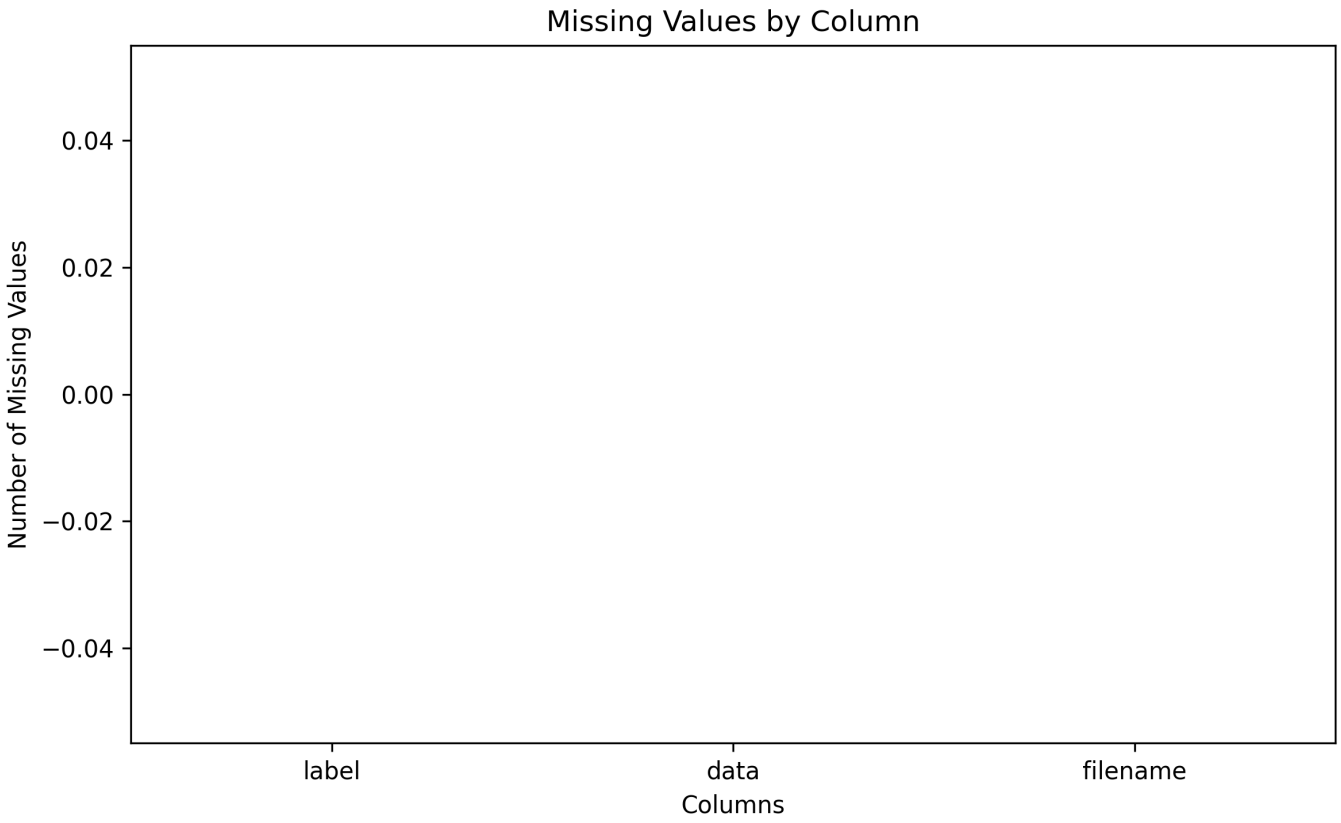
email_length_distribution



label_distribution

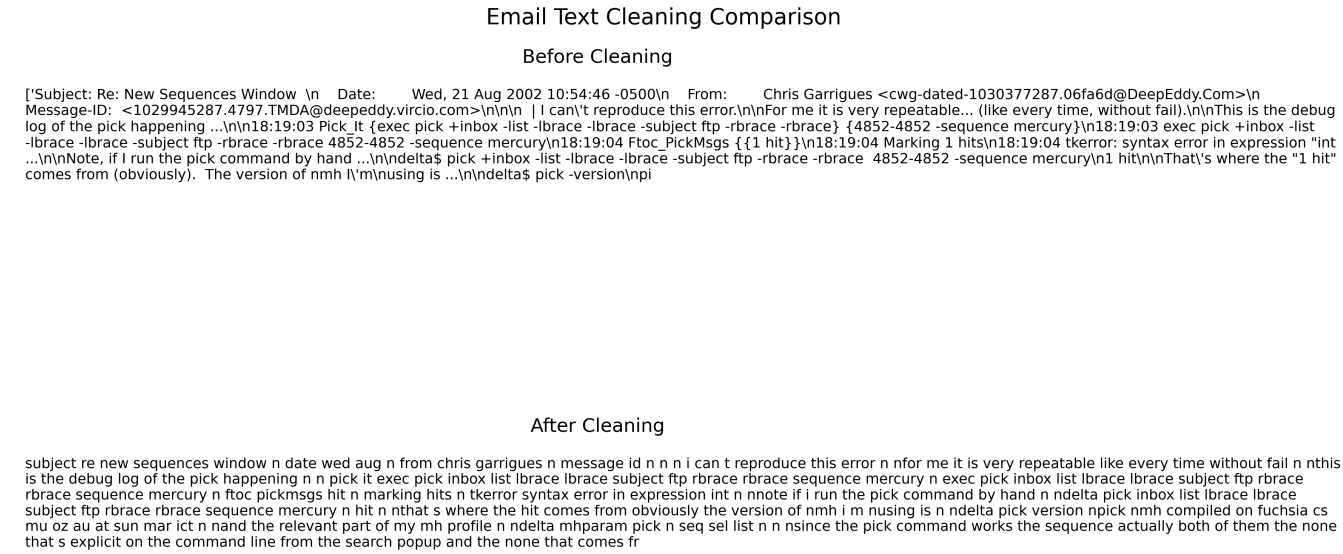


missing_values

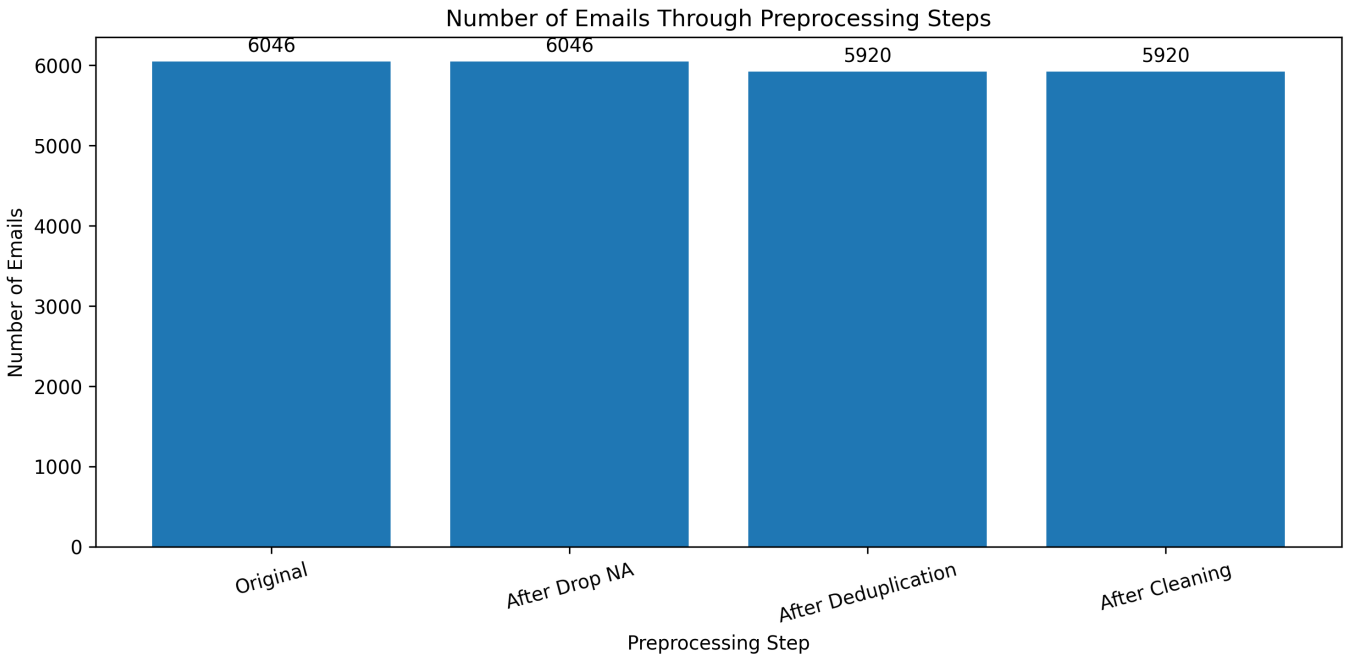


PREPROCESSING

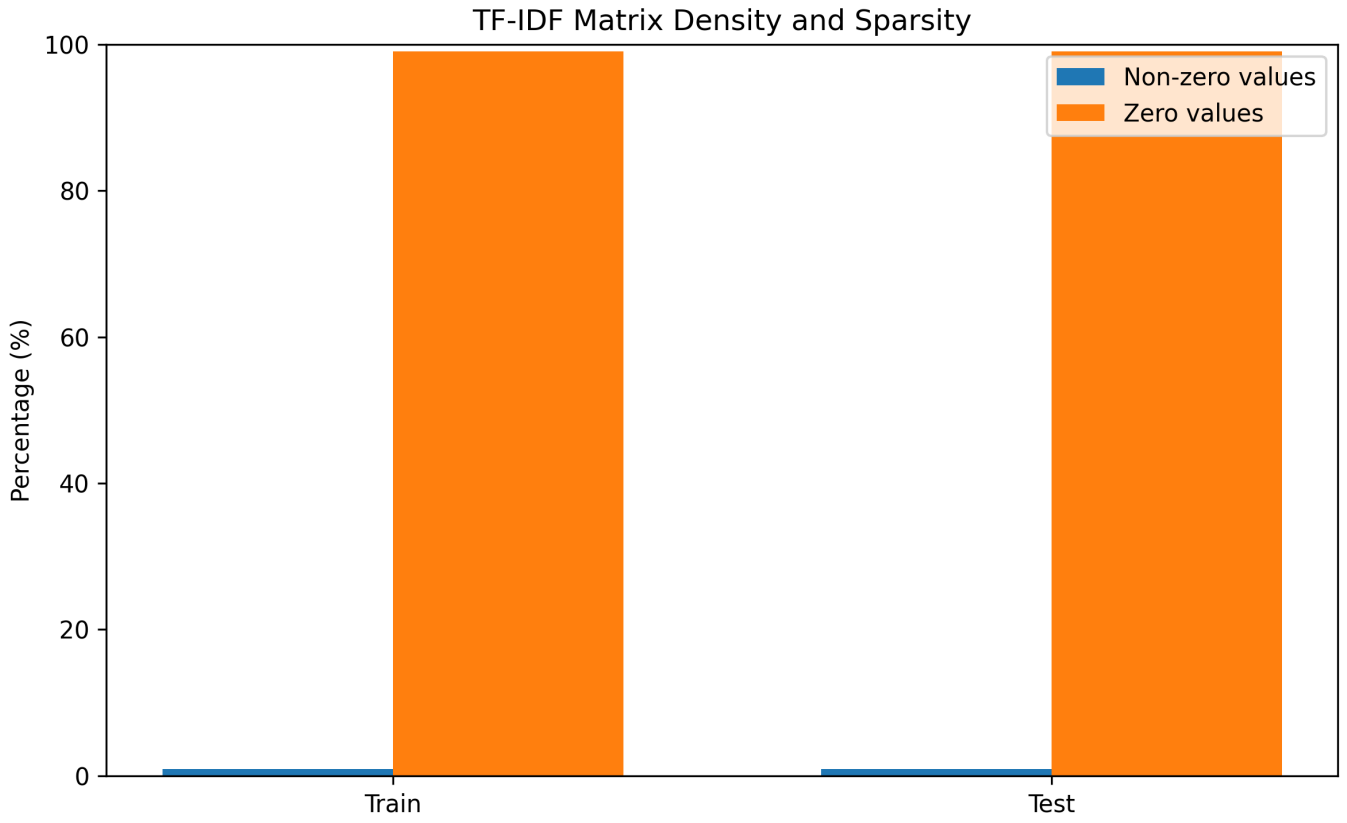
cleaning_comparison



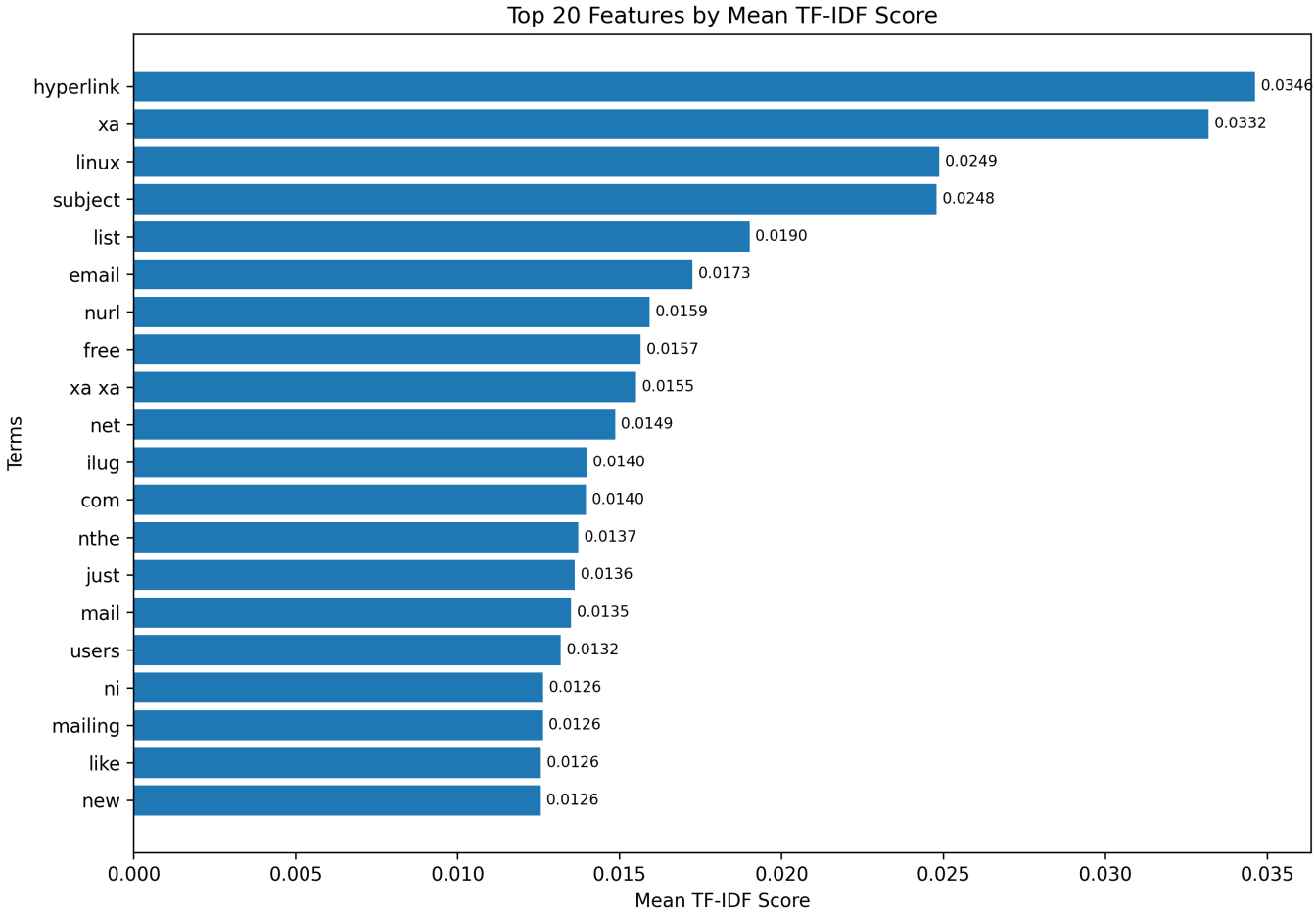
preprocessing_steps



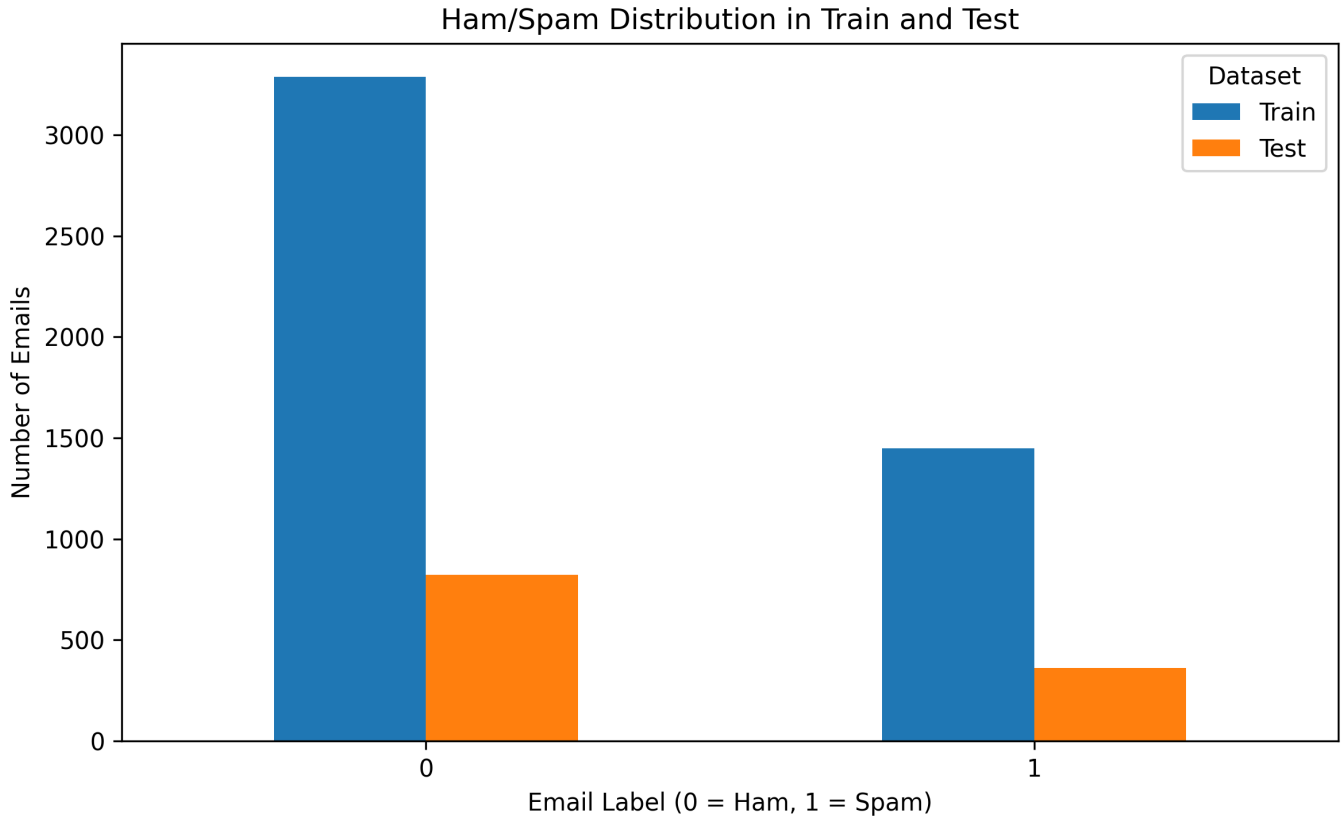
tfidf_sparsity



tfidf_top_features



train_test_distribution



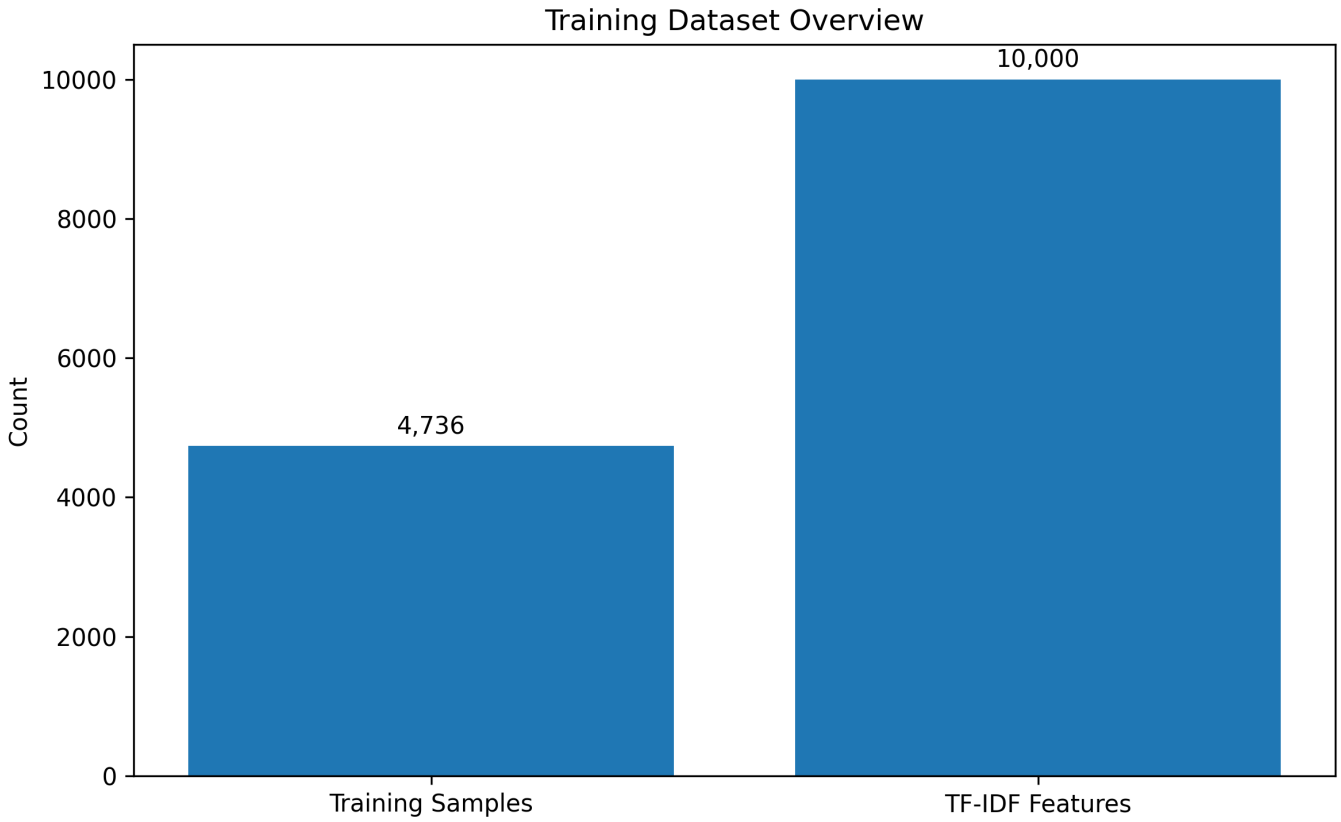
TRAINING

training_configuration

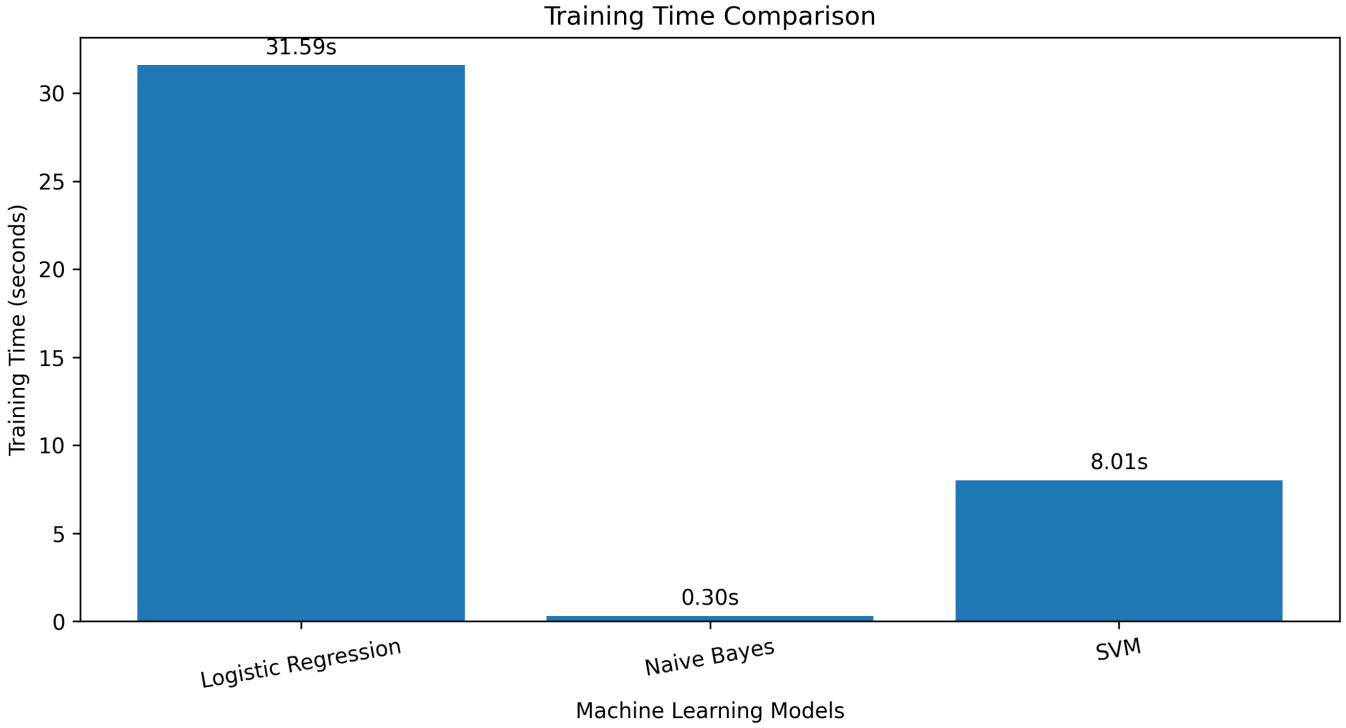
Model Training Configuration

Model	Learning Rate	Iterations	Lambda
Logistic Regression	0.1	1000	N/A
Naive Bayes	N/A	N/A	N/A
SVM	0.01	100	0.01

training_dataset_overview

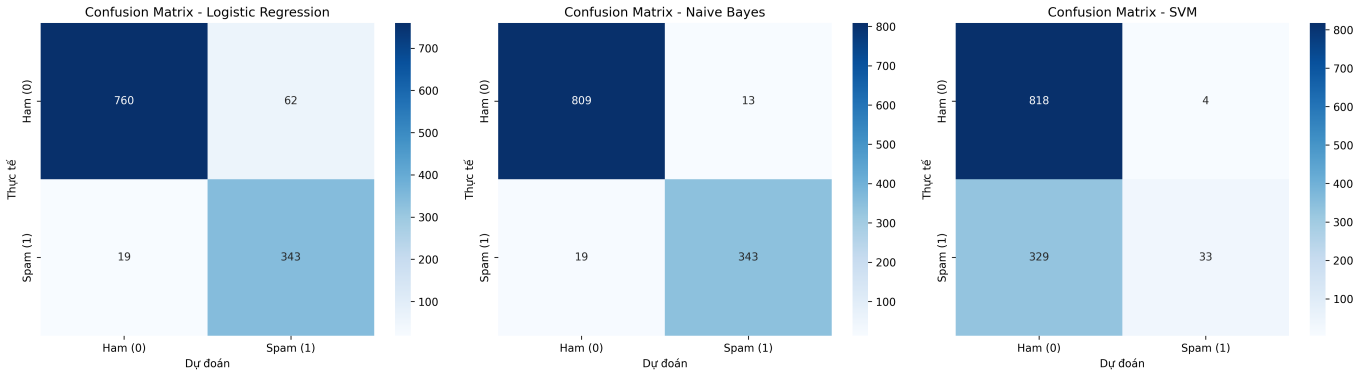


training_time_comparison

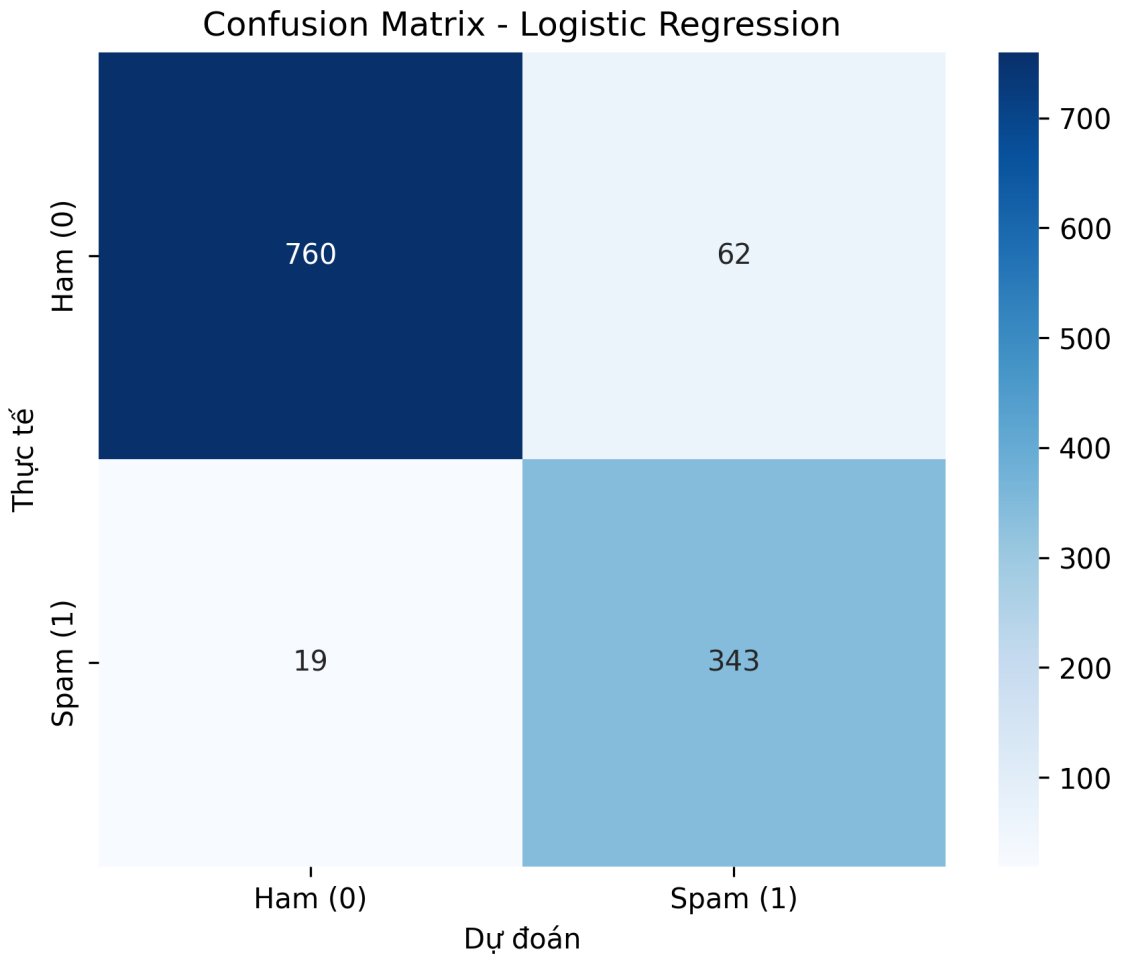


EVALUATION

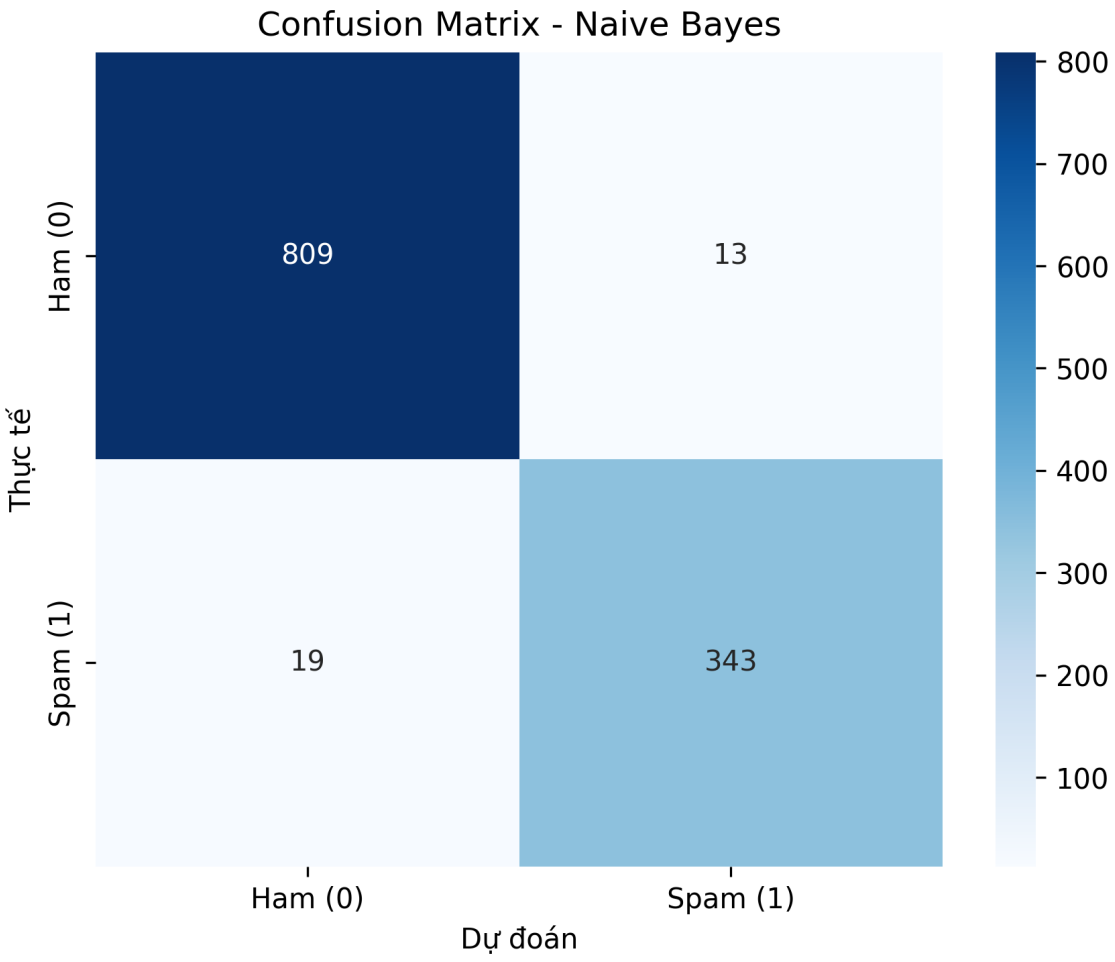
confusion_matrix_all



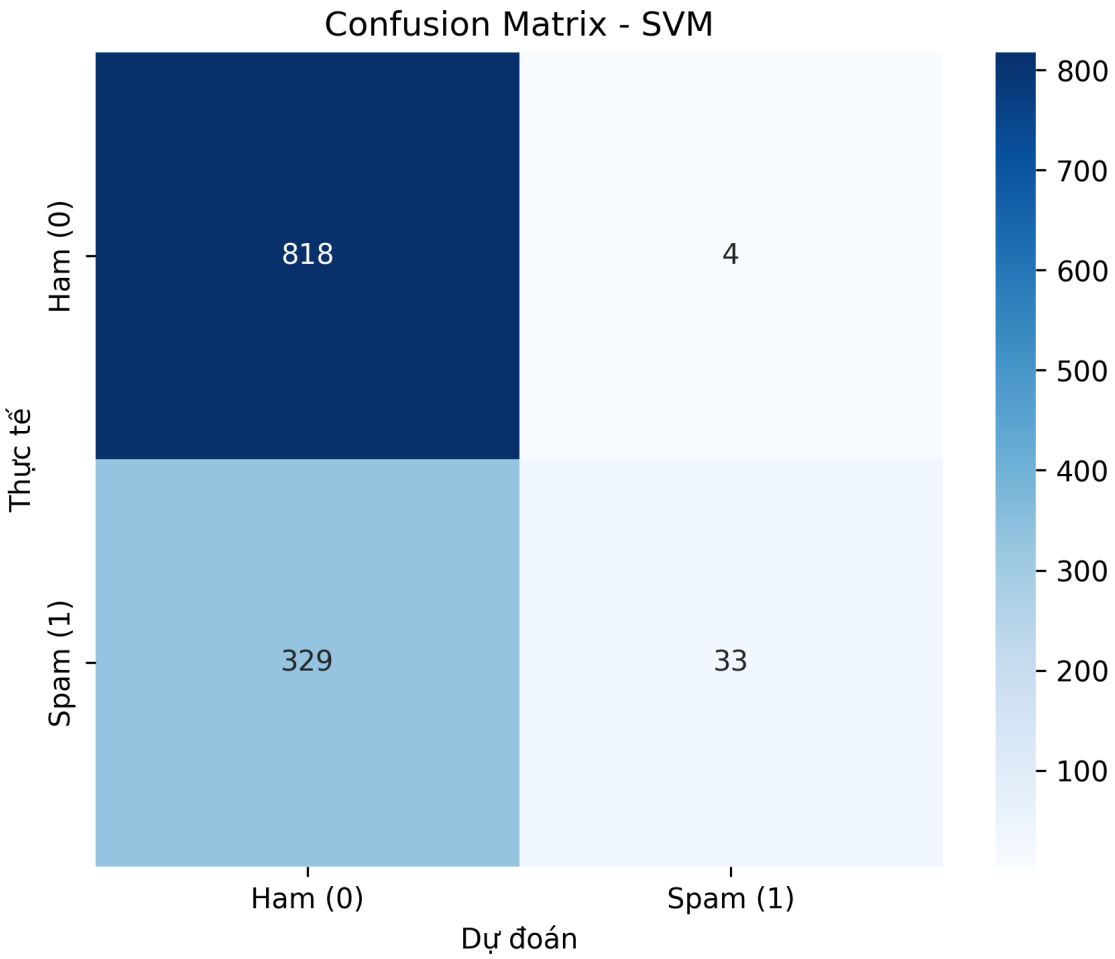
confusion_matrix_lr



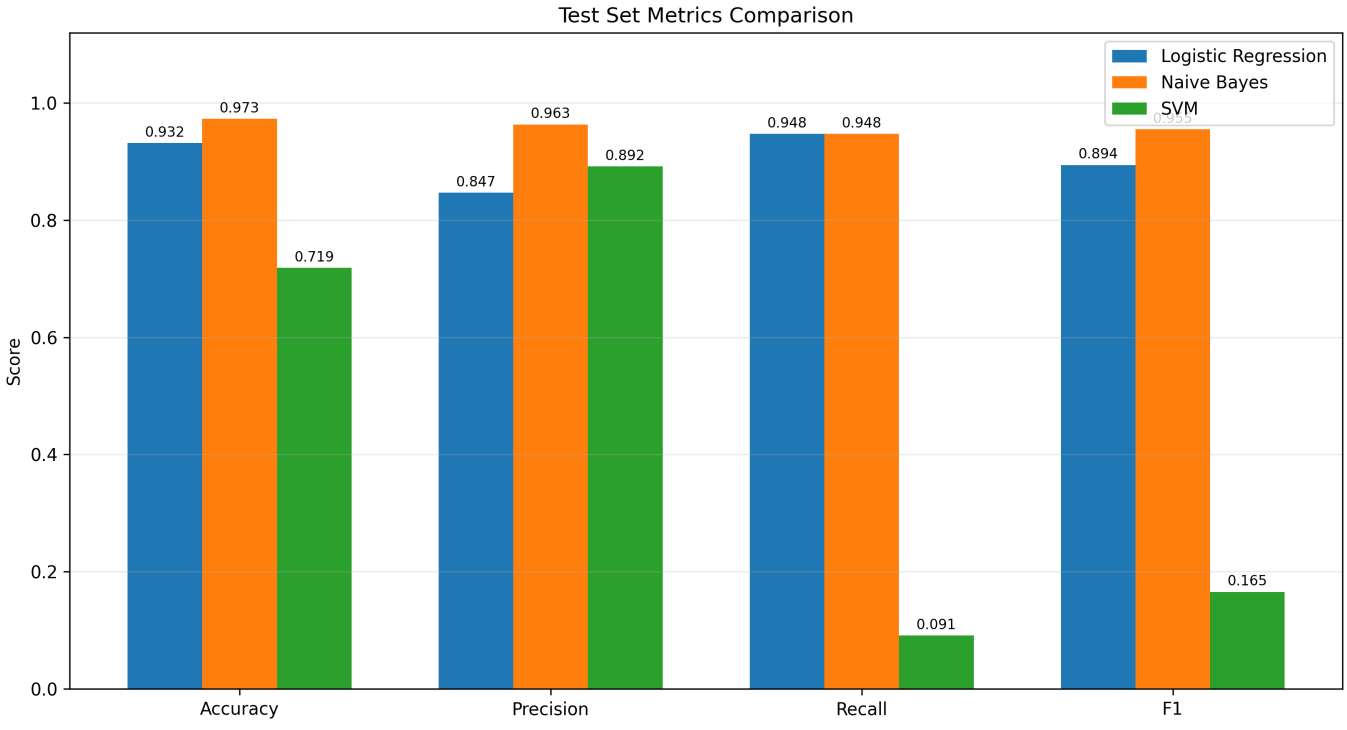
confusion_matrix_nb



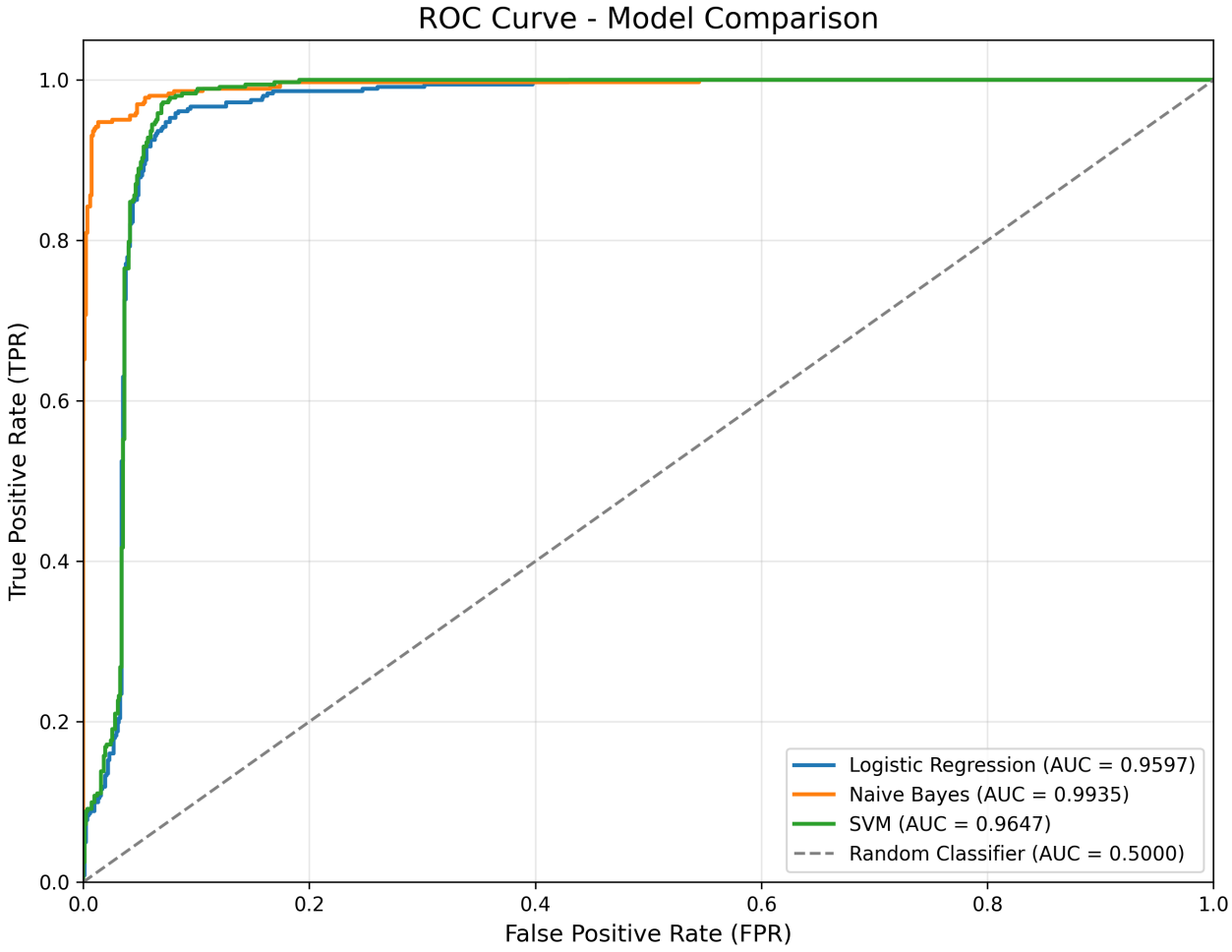
confusion_matrix_svm



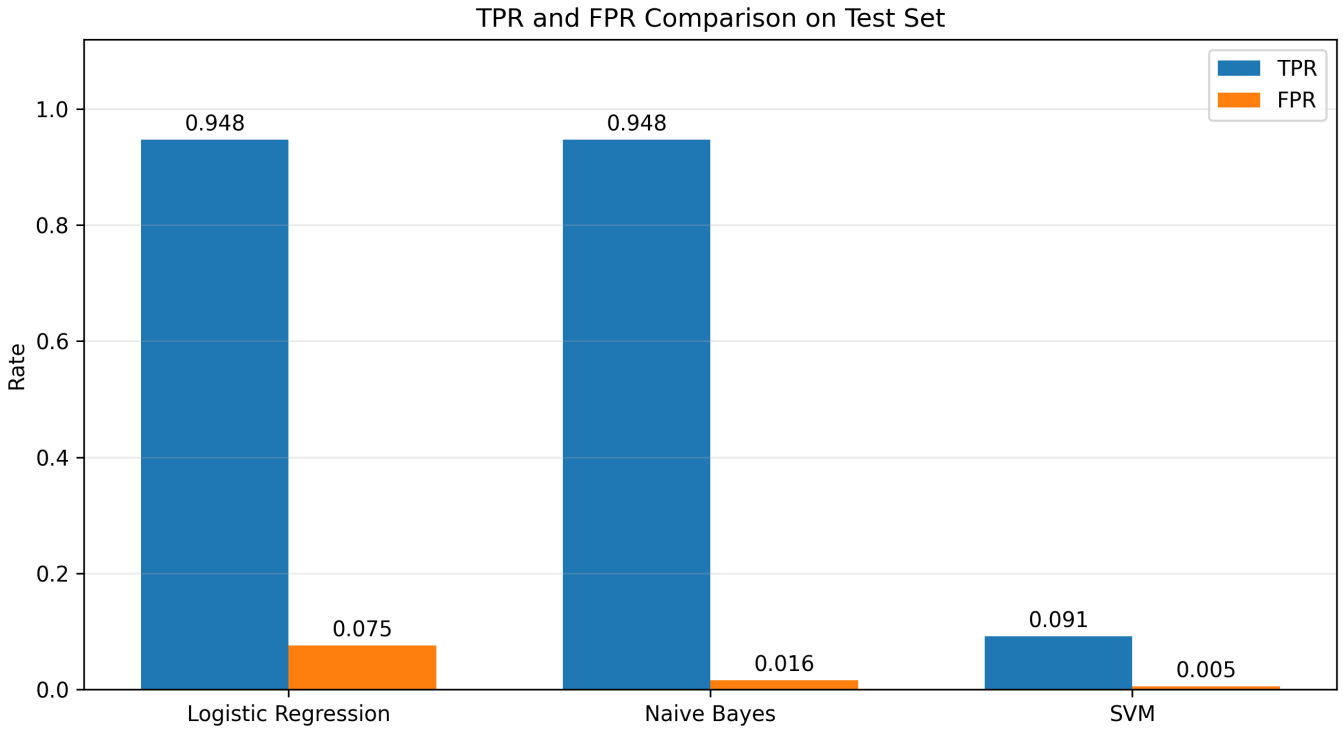
metrics_comparison



roc_curve_all



tpr_fpr_comparison



5. NHAT KY THUC THI

[2026-11-05T01:35:45+07:00] [START] [PIPELINE] Bat dau quy trinh may hoc.

[2026-11-05T01:35:45+07:00] [START] [EDA] Bat dau buoc EDA.

[2026-11-05T01:35:45+07:00] [START] [EDA] Bat dau phan tich du lieu

[2026-11-05T01:35:45+07:00] [SUCCESS] [EDA] Hoan thanh EDA trong 0.6451 giay

[2026-11-05T01:35:45+07:00] [SUCCESS] [EDA] Hoan thanh trong 0.66 giay.

[2026-11-05T01:35:45+07:00] [START] [PREPROCESSING] Bat dau buoc PREPROCESSING.

[2026-11-05T01:35:45+07:00] [START] [PREPROCESSING] Bat dau tien xu ly du lieu

[2026-11-05T01:35:49+07:00] [SUCCESS] [PREPROCESSING] Hoan thanh tien xu ly trong 3.8472 giay

[2026-11-05T01:35:49+07:00] [SUCCESS] [PREPROCESSING] Hoan thanh trong 3.87 giay.

[2026-11-05T01:35:49+07:00] [START] [TRAINING] Bat dau buoc TRAINING.

[2026-11-05T01:35:49+07:00] [START] [TRAINING] Bat dau huan luyen mo hinh

[2026-11-05T01:36:21+07:00] [SUCCESS] [TRAINING - Logistic Regression] Training completed in 31.59 seconds

[2026-11-05T01:36:21+07:00] [SUCCESS] [TRAINING - Naive Bayes] Training completed in 0.30 seconds

[2026-11-05T01:36:29+07:00] [SUCCESS] [TRAINING - SVM] Training completed in 8.01 seconds

[2026-11-05T01:36:30+07:00] [SUCCESS] [TRAINING] Hoan thanh training trong 40.3776 giay

[2026-11-05T01:36:30+07:00] [SUCCESS] [TRAINING] Hoan thanh trong 40.43 giay.

[2026-11-05T01:36:30+07:00] [START] [EVALUATION] Bat dau buoc EVALUATION.

[2026-11-05T01:36:30+07:00] [START] [EVALUATION] Bat dau danh gia cac mo hinh.

[2026-11-05T01:36:30+07:00] [SUCCESS] [EVALUATION - Logistic Regression - Train] Danh gia Logistic Regression tren Train hoan tat. F1=0.9262, AUC=0.9716

[2026-11-05T01:36:30+07:00] [SUCCESS] [EVALUATION - Logistic Regression - Test] Danh gia Logistic Regression tren Test hoan tat. F1=0.8944, AUC=0.9597

[2026-11-05T01:36:31+07:00] [SUCCESS] [EVALUATION - Naive Bayes - Train] Danh gia Naive Bayes tren Train hoan tat. F1=0.9938, AUC=1.0000

[2026-11-05T01:36:31+07:00] [SUCCESS] [EVALUATION - Naive Bayes - Test] Danh gia Naive Bayes tren Test hoan tat. F1=0.9554, AUC=0.9935

[2026-11-05T01:36:31+07:00] [SUCCESS] [EVALUATION - SVM - Train] Danh gia SVM tren Train hoan tat. F1=0.1699, AUC=0.9748

[2026-11-05T01:36:31+07:00] [SUCCESS] [EVALUATION - SVM - Test] Danh gia SVM tren Test hoan tat. F1=0.1654, AUC=0.9647

[2026-11-05T01:36:33+07:00] [SUCCESS] [EVALUATION] Danh gia hoan tat trong 3.27 giay.

[2026-11-05T01:36:33+07:00] [SUCCESS] [EVALUATION] Hoan thanh trong 3.29 giay.

[2026-11-05T01:36:33+07:00] [SUCCESS] [PIPELINE] Hoan thanh toan bo quy trinh trong 48.25 giay.